

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

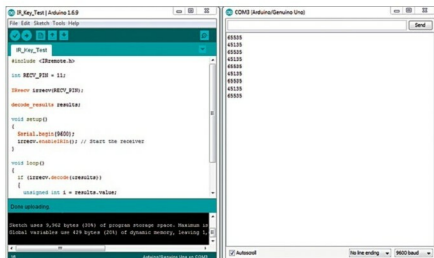


Fig. 5: Example of key configuration



Fig. 6: Screenshot of IR_Final.ino

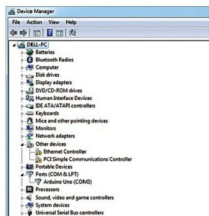


Fig. 7: Device Manager window

following location:

Documents > Arduino > libraries folder

To communicate between Ardui-

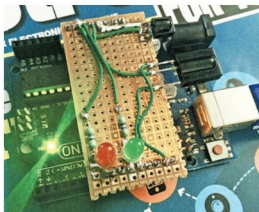


Fig. 8: Author's prototype

no Uno and PC, mode of communication is serial communication. You need to install Remote Volume Control software on the targeted PC, which communicates with Arduino Uno and accordingly performs the task. This software runs in the background and automatically starts when the PC boots.

Initial setup

Step 1. To configure your desired remote control device, install IR-remote library and open IR_Key_Test.ino code from Arduino IDE. Select the correct board from Tools→Board menu in Arduino IDE, and burn the program (sketch) through the standard USB port in your computer.

Step 2. After uploading the code, open Serial Monitor in Arduino IDE.

Point the remote control towards TSOP receiver module and press the key that you want to assign for volume up function continuously. Note the key code number that will repeat again and again when the key is pressed. Screenshot of the example is shown in Fig. 5.

Key code for volume up function as per current serial data is 45135. Note that some remotes send value 65535 or FFFF after sending the key

code. Sometimes, there might be abrupt values due to ambient light or interference, so take five to six readings before assigning the key code. Similarly, repeat the above procedure to find the key codes for volume down and mute functions.

Step 3. After noting down the key codes for all functions, open IR_Final.ino code and write down your key codes as shown in Fig. 6. Save and compile the code.

Testing the project

After uploading IR_Final.ino code to Arduino Uno, connect it to your PC and open Device Manager. Note the COM port assigned to Arduino Uno.

Here, Arduino Uno is assigned COM3 (Fig. 7). Install Remote Volume Control software and start it. Select COM3 in COM Ports menu and select 9600 baud rate from Baud Rate menu. Click on Connect, and if all goes well, volume of the PC will be controlled by your desired remote. The author's prototype of Arduino based Windows PC volume remote control is shown in Fig. 8. **EFY**



Arnab Bansal is an electronics hobbyist and loves to tinker with circuit designs